

MINI-PROJECT SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)
Mini-Project Title: Mini-Project 3 — OCR Expense Tracker & Receipt Parser (Flutter & Dart)
Team / Student Name: Nguyễn Phan Nhật Quang
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1. GENERAL INFORMATION & DELIVERABLE LINKS

- **Team Members:**
 - Nguyễn Phan Nhật Quang — Student ID: 23IT.B176 — Role: Individual project (architecture, OCR & parser, database, UI/charts, testing) — Contribution: 100%
 - 🔗 **Live Demo URL (Release APK):** [app-release.apk](#) (universal) · [arm64-v8a build, 31.9 MB](#) · [Release page](#)
 - 📁 **GitHub Repository:** https://github.com/nquang510/miniproject_orc
- Tech stack:** Flutter 3.47.6 · Dart 3.13 · `camera` · `google_mlkit_text_recognition` (on-device Latin model) · `sqflite` · `image` · `CustomPainter` (no chart library).

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required Feature	Status	Implementation Details & Acceptance Level
1	Camera capture & image cropping	✅ Complete	Live <code>CameraPreview</code> with a flash cycle (off / auto / on / torch), tap-to-focus (<code>setFocusPoint</code> + <code>setExposurePoint</code>) with an animated focus ring, and a receipt framing overlay drawn by <code>CropOverlayPainter</code> . The still photo is cropped to the frame in a background isolate. The camera is released and re-acquired on app lifecycle changes. Gallery import and manual entry are fallbacks.
2	On-device text recognition	✅ Complete	<code>google_mlkit_text_recognition</code> (bundled Latin model, works offline, no cloud cost). ML Kit's time is measured with a <code>Stopwatch</code> and shown on the review screen. OCR runs on the cropped image, downscaled to ≤1600 px, to keep latency low.
3	Regex heuristic parser	✅ Complete	Reads totals in 150,000 VND, 150.000 đ, 1.250.000, 150,000.00 and 45k formats; dates in DD/MM/YYYY, DD-MM-YY, YYYY-MM-DD and Ngày ... tháng ... năm ...; merchant names from 39 known brands or from the receipt header. Each line is scored (total keywords +8, currency mark +2, cash/change/VAT/discount lines −7). Diacritics are folded first, and O-for-0 OCR errors are repaired. 16 unit tests.
4	Interactive review screen	✅ Complete	Every field can be edited. Fields carry a confidence hint (high / medium / low, with low highlighted). All detected amounts appear as tap-to-use chips, and the raw OCR text is expandable. Amounts are formatted as you type (150.000 đ). Leaving an unsaved scan asks for confirmation and deletes its images.
5	Local database & categories	✅ Complete	<code>sqflite</code> table <code>expenses</code> with indexes on <code>date</code> and <code>category</code> . The five categories (Food, Study, Travel, Gear, Entertainment) are suggested automatically from brand and keyword matches. Supports create, edit, and delete with undo.
6	Receipt thumbnail caching	✅ Complete	The cropped JPEG and a 240 px thumbnail are saved in <code><app documents>/receipts/</code> . The database stores relative file names. List tiles decode at display size (<code>cacheWidth</code>).
7	CustomPainter charts	✅ Complete	Donut: sweeps in clockwise; tapping a slice (hit-tested by angle) thickens it and fades the others; legend is linked; Week / Month / All filter. Weekly bar chart: staggered bar growth, "nice" Y-axis steps, dashed grid, tooltip on tap, previous/next week navigation. No third-party chart package.
8	Extras	✅ Complete	History grouped by day with search and category filter, swipe-to-delete with undo, light/dark theme, 25 automated tests, <code>flutter analyze</code> with no issues.

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

lib/	
├─ core/	AppScope (InheritedNotifier DI), theme, VND formatters, date helpers
├─ models/	Expense, ExpenseCategory

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├─ data/      SqfliteExpenseRepository (+InMemory for tests), ReceiptImageStore
├─ services/  OcrService, receipt_layout (row merge), ReceiptParser, receipt_scanner
├─ state/     ExpenseStore (ChangeNotifier) + chart aggregates
└─ ui/        screens (home, dashboard, history, camera, review), widgets (painters)

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Data flow. Camera → ReceiptImageStore.saveReceipt (isolate: EXIF rotation → crop to frame → resize → thumbnail) → OcrService (ML Kit, then lines merged into rows) → ReceiptParser (pure Dart) → draft Expense → ReviewScreen → ExpenseStore.add → SQLite.

State management. One ExpenseStore (ChangeNotifier) is exposed through an InheritedNotifier (AppScope), so no extra package is needed. Writes go to the repository first, and the in-memory list changes only after the write succeeds. The chart aggregates (totalsByCategory, dailyTotals) are computed from that list.

Testability. The parser and layout logic never touch ML Kit types. They work on plain strings and boxes, so every heuristic has a unit test. The repository is an interface, and an in-memory version drives the widget tests and screenshot rendering.

Exception handling.

- Camera permission denial and device errors (CameraException) show an explanation with *Try again* and *Pick from gallery*.
- Focus and flash calls that a device does not support are ignored.
- If OCR fails after the image was saved, the image is deleted and an error snackbar is shown.
- An empty OCR result shows a “No text detected” card instead of empty fields.
- Database load errors are kept in the store, and save errors appear in a snackbar.

4. EMPIRICAL EVIDENCE & SCREENSHOTS

Screenshots below were rendered from the real app widgets at phone size (1080×2340) by `screenshot_test/`, using sample data. On the camera screen the live feed is a simulated receipt photo fed through a fake `CameraPlatform`. Replace them with device captures from the demo video if required.

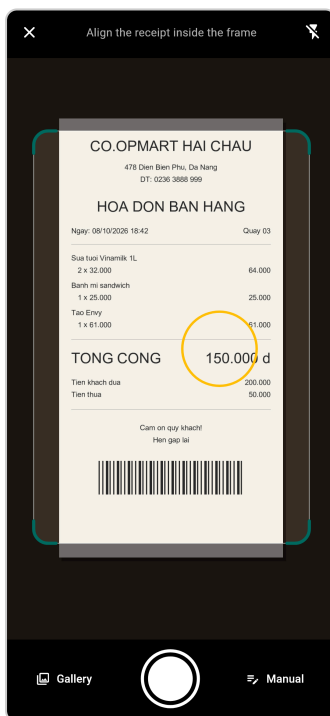


Fig. 1 – Capture. Framing overlay with corner brackets, tap-to-focus ring, flash toggle (top right), gallery / manual fallbacks.

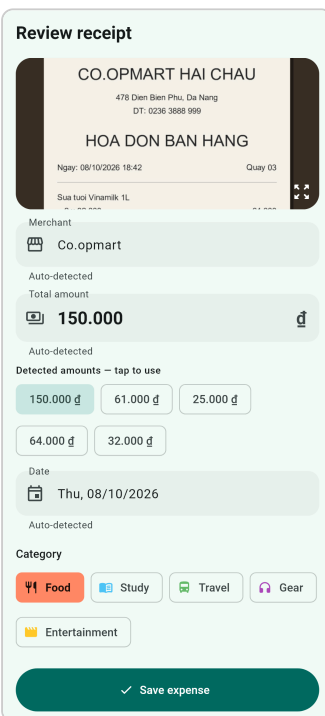


Fig. 2 – Review. Image cropped to the frame. Parser output: merchant *Co.opmart*, total *150.000 đ* (cash 200.000 and change 50.000 rejected), date 08/10/2026, category Food; other amounts as chips.

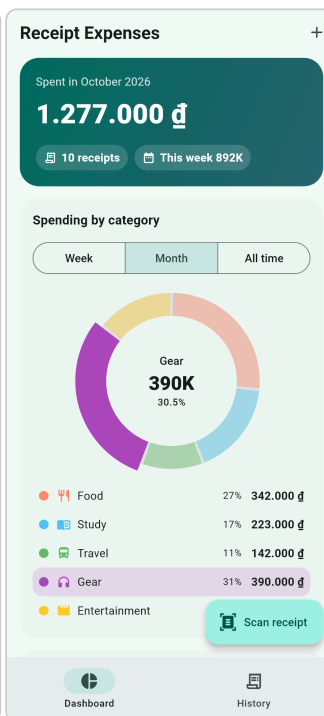


Fig. 3 – Donut chart. “Gear” slice selected by tap: it thickens, the others fade, and the centre and legend update.

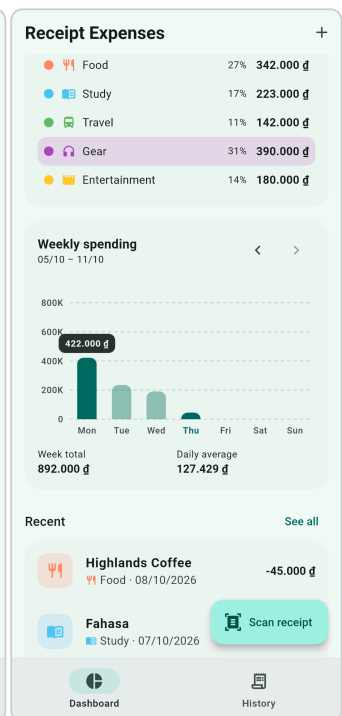


Fig. 4 – Weekly bar chart. Tooltip on the tapped bar, today (Thu) highlighted, “nice” axis steps, week navigation.

Verification results

- `flutter analyze` → *No issues found*. `flutter test` → **25/25 passed**: 16 parser tests (Vietnamese/English receipts, VAT-inclusive totals, label/value on separate rows, invalid and future dates, O-for-0 repair), plus chart geometry, store aggregates, and donut/bar tap interaction.
- `flutter build apk --release` → `app-release.apk` (85.3 MB universal). With `--split-per-abi`: arm64-v8a 31.9 MB, armeabi-v7a 25.6 MB. Most of the size is the bundled ML Kit model, which is what lets OCR work offline.

5. TECHNICAL CHALLENGES & RESOLUTIONS

1. ML Kit splits a receipt's columns into separate text blocks. ML Kit groups text by layout block. On a two-column receipt, "TỔNG CỘNG" sits in the left block and "150.000" in the right block, so they come back far apart, and a line-based regex cannot link the label to its value. *Resolution*: `mergeIntoRows` sorts all ML Kit lines by their vertical centre. A line joins the current row when its centre is within half a line height of the row, which tolerates slight camera tilt. Each row is then joined left to right, giving "TỔNG CỘNG 150.000" before parsing. When a label still ends up on its own row, the parser carries the keyword score over to the next row. Both cases are unit-tested.

2. Choosing the real total among many numbers. A receipt holds item prices, VAT, cash given (Tiền khách đưa 200.000) and change, as well as phone numbers, tax IDs and dates. Vietnamese receipts also use `.` and `,` as thousands separators, and OCR often reads 0 as o. *Resolution*: the parser runs in stages:

1. Fold diacritics and repair O-for-0 errors.
2. Remove dates, times and percentages; ignore phone and tax-ID lines.
3. Read number tokens by digit grouping, so `150.000` = `150,000` = 150000 and `150,000.00` keeps its decimal.
4. Score each candidate (total keyword +8, subtotal +2, currency mark +2, cash/change/discount/VAT -7, lower half of the receipt +1).

"đã gồm VAT" is not treated as a tax line. If no line scores high enough, the largest non-cash amount is used and marked low confidence for the user to check.

3. Crop mismatch between preview and photo; release-build issues. The preview stream and the still photo can have different aspect ratios, so cropping the photo with the on-screen frame fractions cut off part of the receipt. *Resolution*: the photo is first centre-cropped to the preview's aspect ratio (what the user actually saw), then the frame fractions are applied. This runs in a `compute` isolate, because decoding a 12 MP JPEG in Dart takes about a second and would freeze the UI. Two build issues came up for the release APK:

- On Windows, Gradle failed with "Could not close incremental caches" in `camera_android_camerax` and `image_picker_android`, because the project (drive E:) and the pub cache (drive C:) are on different drives. Setting `kotlin.incremental=false` in `android/gradle.properties` fixed it.
- The ML Kit plugin references the Chinese, Japanese, Korean and Devanagari recognisers only as `compileOnly` dependencies. With R8 minification turned on, `-dontwarn` rules for those packages were added to `proguard-rules.pro` so the shrinker does not stop on the missing classes.